Algorithms 800 - Homework 1

due Thursday, March 21, 2013

Reading Chapters 1 through 4

1. Consider algorithms A_1, A_2, A_3, and A_4 that have running times of $f_1(n) = 250000 \log_{10} n$, $f_2(n) = n^3/15 + n^2$, $f_3(n) = 40n^2 + n$, and $f_4(n) = 2^n/10000$, respectively (in microseconds). Draw a table showing the size of a problem that can be solved in 1 second, 1 hour and 1 year by each algorithm. It is similar to, but different from the tables on page 15 of CLRS, or a handout.

Hint: You don't have to solve it analytically; you can write a simple program to obtain the approximate answers. Give at least two most significant digits of the answers.

2. Prove or disprove the following statements:
a) $n^4 = \Omega(n \log n + n^4)$
b) $n^2 - n = \Theta(n^3)$
c) $n! = \Theta(2^n)$
d) $\log_{10} n = \Theta(\lg \sqrt{n})$

3. Solve problems 3.4/aceg page 62. Justify your answers.

4. Solve exercises 4.5-1/abcd page 96. Justify your answers.

5. Solve problems 4.1/aceg page 107. Justify your answers. In this exercise extra calculus insight is welcome. Good estimates are good enough.

6. Find the number of comparisons performed by the MAXMIN algorithm for the number of elements in the set S equal to 7, 8, 9 and 18. Show the details of the computation for $n = 13$. The answer for 0, 1, 2 is 0, 0, 1, respectively.